GRAPHIC ERA (DEEMED TO BE UNIVERSITY)

Department of Computer Science and Engineering

TCS-392: INTRODUCTION TO CRYPTOGRAPHY

UNIT–I ASSIGNMENT

Security Fundamentals and Classical Cryptography

Name: _____ **University Roll No.:** _____

Section: _____ **Date of Submission:** _____

Instructions

- Attempt all questions.
- Show all steps in numerical problems.
- Unless otherwise stated, use $A = 0, B = 1, \ldots, Z = 25$.
- Draw neat diagrams wherever required.
- Answers should include justification and not only final results.

Assignment Questions

Q1. Explain the CIA triad in detail. Differentiate between data confidentiality and privacy, data integrity and system integrity, and also explain authenticity and accountability with suitable examples.

Q2. Explain the security impact levels **Low, Moderate, and High**. Classify the following assets according to confidentiality, integrity, and availability requirements:

- Student grades
- Examination question papers
- Public university website
- Hospital allergy records
- Faculty directory

Q3. Define and differentiate among:

(a) Threat

(b) Vulnerability

(c) Attack

(d) Security Service

(e) Security Mechanism

Explain their relationship using a suitable example.

Q4. Explain passive and active attacks. Discuss:

(a) Release of message contents

(b) Traffic analysis

(c) Masquerade

(d) Replay

(e) Message modification

(f) Denial of service

Q5. Prepare a detailed comparison between passive and active attacks in terms of objective, data modification, detectability, security objectives affected, and typical countermeasures.

Q6. Explain the major X.800 security services:

(a) Peer-entity authentication

(b) Data-origin authentication

(c) Access control

(d) Data confidentiality

(e) Data integrity

(f) Nonrepudiation

(g) Availability

Q7. Explain the following specific and pervasive security mechanisms:

(a) Encipherment

(b) Digital Signature

(c) Authentication Exchange

(d) Traffic Padding

(e) Routing Control

(f) Notarization

(g) Security Audit Trail

(h) Security Recovery

Q8. Draw and explain the **General Model for Network Security**. Clearly describe the role of sender, recipient, opponent, trusted third party, secret information, information channel, and security-related transformation.

Q9. Explain the four basic tasks involved in designing a security service:

(1) Designing the transformation algorithm

(2) Generating secret information

(3) Distributing secret information

(4) Designing the communication protocol

Q10. Explain the **Network Access Security Model**. Differentiate between information-access threats and service threats. Also explain gatekeeper functions and internal security controls.

Q11. Define the following cryptographic terms:

(a) Plaintext

(b) Ciphertext

(c) Encryption

(d) Decryption

(e) Cipher

(f) Cryptography

(g) Cryptanalysis

(h) Cryptology

Q12. Explain the symmetric cipher model and its five basic ingredients. Also explain why the secrecy of the key is more important than secrecy of the algorithm.

Q13. Explain the three dimensions used to characterize cryptographic systems:

(a) Type of transformation

(b) Number of keys

(c) Method of processing plaintext

Differentiate between substitution and transposition, symmetric and asymmetric encryption, and block and stream ciphers.

Q14. Explain cryptanalysis and brute-force attack. Discuss ciphertext-only, known-plaintext, chosen-plaintext, chosen-ciphertext, and chosen-text attacks.

Q15. A cipher uses an n-bit key. Derive:

$$\text{Total Key Space} = 2^n$$

and calculate the total and average number of key trials for:

$$n = 8, \quad 16, \quad 32, \quad 56, \quad 128.$$

Q16. Caesar Cipher Numerical: Encrypt the plaintext

`CRYPTOGRAPHY`

using key

$$k = 7.$$

Show the numerical representation and every modulo-26 step.

Q17. Caesar Cipher Numerical: Decrypt

`YMNXNXFQJ`

using key

$$k = 5.$$

Also determine whether the resulting plaintext is meaningful.

Q18. A Caesar cipher transforms plaintext value $P = 19$ into ciphertext value $C = 4$. Determine the key. Then verify your answer using the encryption formula:

$$C = (P + k) \bmod 26.$$

Q19. Explain why monoalphabetic substitution cipher has a key space of:

$$26!$$

and yet remains vulnerable to frequency analysis. Explain the role of single-letter frequencies, digrams, trigrams, and repeated patterns.

Q20. Consider the substitution alphabet:

P	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
C	Q	W	E	R	T	Y	U	I	O	P	A	S	D	F	G	H	J	K	L	Z	X	C	V	B	N

Encrypt:

`NETWORK`

and decrypt:

`LTEXKOZN`.

Q21. Playfair Cipher Numerical: Construct the Playfair matrix using the keyword:

`MONARCHY`

and encrypt:

`BALLOON`.

Show the preprocessing of repeated letters, formation of digrams, and the Playfair rule used for each pair.

Q22. Using the Playfair matrix generated from `MONARCHY`, encrypt the digrams:

$$AR, \qquad MU, \qquad HS, \qquad EA.$$

For each digram, specify whether it uses the same-row rule, same-column rule, or rectangle rule.

Q23. Hill Cipher Numerical: Consider the key matrix:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}.$$

Encrypt the plaintext digram:

`HI`

using:

$$C = PK \pmod{26}.$$

Q24. For the matrix:

$$A = \begin{bmatrix} 5 & 8 \\ 17 & 3 \end{bmatrix},$$

calculate:

(a) $\det(A)$

(b) $\det(A) \bmod 26$

(c) Multiplicative inverse of the determinant modulo 26

(d) $A^{-1} \bmod 26$

Hence explain why invertibility of the key matrix is essential in Hill cipher.

Q25. Vigenère Cipher Numerical: Encrypt:

`ATTACKATDAWN`

using the keyword:

`LEMON`.

Repeat the keyword to match the plaintext length and show all numerical calculations.

Q26. Decrypt the Vigenère ciphertext:

`LXFOPVEFRNHR`

using keyword:

`LEMON`.

Also explain why the Vigenère cipher is more resistant to simple frequency analysis than a monoalphabetic cipher.

Q27. Vernam and One-Time Pad: For:

$$P = 101101101001$$

and

$$K = 011010110111,$$

calculate:

$$C = P \oplus K.$$

Then decrypt the ciphertext using:

$$P = C \oplus K.$$

Explain the conditions under which the One-Time Pad provides perfect secrecy.

Q28. Transposition Cipher Numerical: Encrypt:

`MEETMEAFTERTHETOGAPARTY`

using a Rail Fence cipher of depth 2. Show the complete rail arrangement and final ciphertext.

Q29. Columnar Transposition Numerical: Encrypt:

`ATTACKPOSTPONEDUNTILTWOAMXYZ`

using the key:

$$4 \quad 3 \quad 1 \quad 2 \quad 5 \quad 6 \quad 7.$$

Show:

(a) Matrix formation

(b) Column labels

(c) Column reading order

(d) Final ciphertext

Then explain why double transposition improves security.

Q30. Explain the working of a rotor machine. Calculate the total number of substitution states for machines containing:

$$3, \quad 4, \quad 5$$

rotors, where each rotor has 26 positions. Also explain the odometer effect.

Q31. Explain steganography and differentiate it from cryptography. Discuss character marking, invisible ink, pin punctures, and modern LSB-based steganography. An RGB image has resolution:

$$1920 \times 1080$$

and one LSB from each of the three RGB channels is used. Calculate:

(a) Total number of pixels

(b) Hidden bits per pixel

(c) Total hidden bits

(d) Total hidden bytes

(e) Approximate storage capacity in MB

— End of Unit–I Assignment —